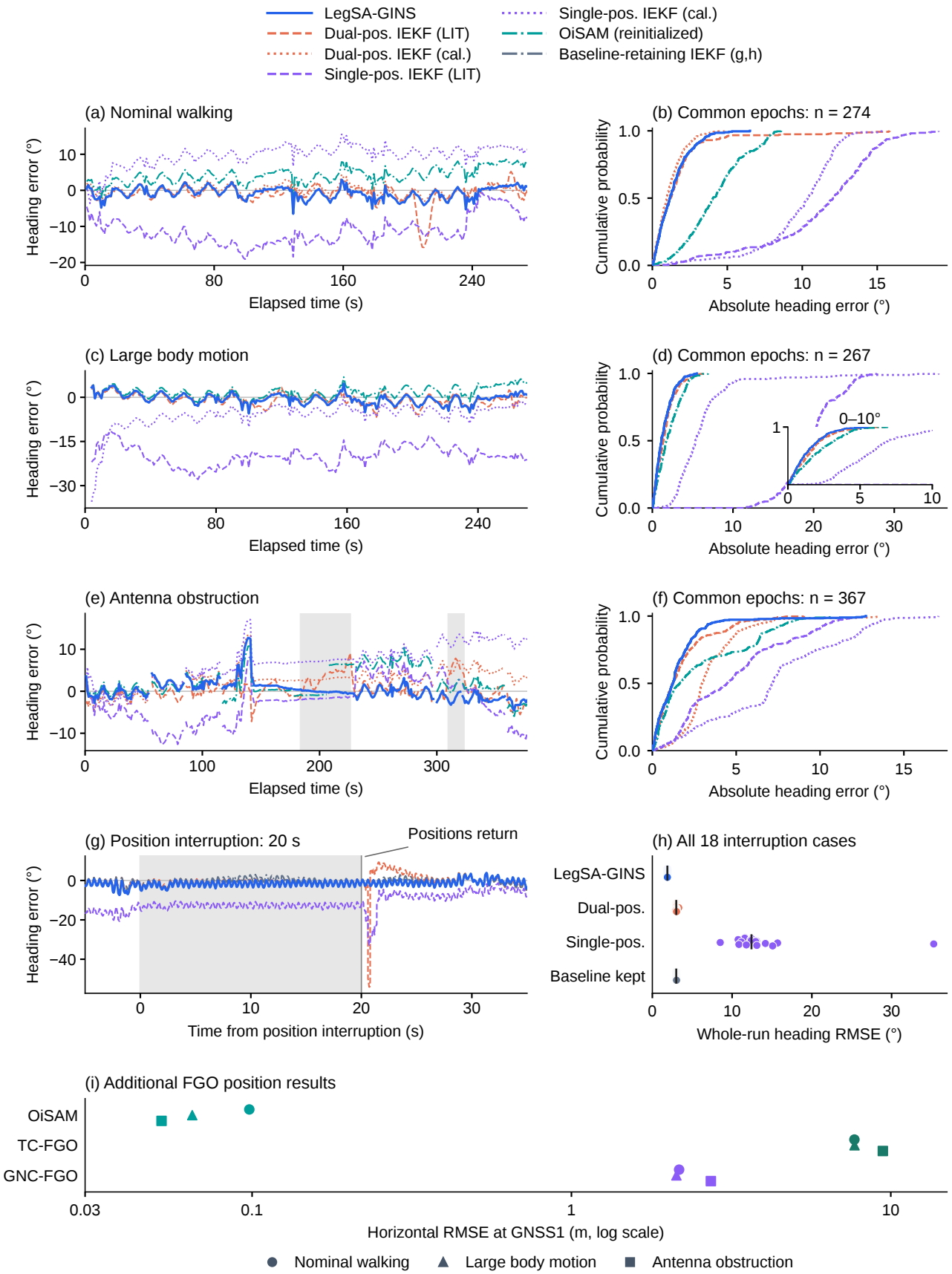

LegSA-GINS
Dual-pos. IEKF (LIT)
Dual-pos. IEKF (cal.)
Single-pos. IEKF (LIT)
Single-pos. IEKF (cal.)
OiSAM (reinitialized)
Baseline-retaining IEKF (g,h)
(a) Nominal walking
Heading error (°)
Elapsed time (s)
(b) Common epochs: n = 274
Cumulative probability
Absolute heading error (°)
(c) Large body motion
(d) Common epochs: n = 267
0–10°
(e) Antenna obstruction
(f) Common epochs: n = 367
(g) Position interruption: 20 s
Positions return
Time from position interruption (s)
(h) All 18 interruption cases
LegSA-GINS
Dual-pos.
Single-pos.
Baseline kept
Whole-run heading RMSE (°)
(i) Additional FGO position results
OiSAM
TC-FGO
GNC-FGO
Horizontal RMSE at GNSS1 (m, log scale)
Nominal walking
Large body motion
Antenna obstruction